Huawei Access Node HSR-ACC-74 Optical Transport Failure Recovery Procedure

Node: HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2958

Description

The Huawei Access Node HSR-ACC-74 is a critical component of the HSR Layout Access site's optical transport network. As such, it is susceptible to failure due to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This is primarily due to the node's reliance on optical transport protocols such as OTN (Optical Transport Network) or SDH (Synchronous Digital Hierarchy) for high-speed data transmission over fiber optic cables. The OTN protocol, in particular, relies on the correct functioning of SFP/XFP modules and fiber links to ensure accurate transmission of optical signals. If these components fail, the node may report a critical alarm, such as ALARM_2958, indicating a severe issue with the optical transport mechanism. To prevent data loss and network downtime, it is essential to quickly identify the root cause of the failure and perform prompt repairs.

Pre-requisite Checks

- Dispatch remote hands to the HSR Layout Access site to ensure physical access to the node.

- Verify the status of redundant links to ensure the network remains operational in case of a single link failure.

- Review the node's alarm history to determine the exact timing and cause of the ALARM_2958 alarm.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device, Expected Output: Device information, including SFP/XFP status and optical signal quality, Description: This command displays detailed information about the node's devices, including SFP/XFP status and optical signal quality. This will help isolate issues related to SFP/XFP degradation or fiber cuts.

Command: display optical-power, Expected Output: Optical power levels for each fiber link, Description: This command displays the current optical power levels for each fiber link, helping to identify any issues with Rx/Tx power loss.

Command: display alarm-alarm, Expected Output: Detailed alarm information, including alarm code and description, Description: This
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command displays detailed information about the ALARM_2958 alarm,
including the alarm code and description.
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Resolution Steps

Step 1: Step: Replace faulty SFP/XFP module, Command: replace device {SFP/XFP Slot}, Description: The faulty SFP/XFP module must be replaced with a new one to restore normal operation of the node. The {SFP/XFP Slot} should be replaced with the actual slot number where the faulty module is located.

Step 2: Step: Repair or replace fiber cable, Command: replace optical-fiber {Fiber Link}, Description: If a fiber cut is identified, the affected fiber cable must be repaired or replaced to restore normal operation of the node. The {Fiber Link} should be replaced with the actual link number where the fiber cut occurred.

Step 3: Step: Adjust Rx/Tx power levels, Command: adjust optical-power {Rx/Tx Port}, Description: If Rx/Tx power loss is identified, the power levels must be adjusted to restore normal operation of the node. The {Rx/Tx Port} should be replaced with the actual port number where the power loss occurred.

Post-Check Verification

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Command: display device, Expected Output: Device information,
including SFP/XFP status and optical signal quality, Description: This
command verifies that the SFP/XFP module has been successfully
replaced or adjusted to restore normal operation of the node.

Command: display optical-power, Expected Output: Optical power levels
for each fiber link, Description: This command verifies that the fiber
cable has been successfully repaired or replaced, and that Rx/Tx power
levels are within normal operating limits.

Command: display alarm-alarm, Expected Output: No ALARM_2958 alarm,
Description: This command verifies that the ALARM_2958 alarm has been
cleared, indicating that the node is operating normally.
```

Roll-back Procedures

- Step: Revert SFP/XFP replacement, Command: undo replace device {SFP/XFP Slot}, Description: If the SFP/XFP replacement fails, the original module must be reverted to its previous state to avoid further network disruptions.

- Step: Revert fiber cable repair/replacement, Command: undo replace optical-fiber {Fiber Link}, Description: If the fiber cable repair or replacement fails, the original cable must be reverted to its previous state to avoid further network disruptions.

- Step: Revert Rx/Tx power level adjustment, Command: undo adjust optical-power {Rx/Tx Port}, Description: If the Rx/Tx power level adjustment fails, the original power levels must be reverted to their previous state to avoid further network disruptions.